Solving Literal Equations — Set A

Easy level · 30 problems · A worked example comes before each type. Show every step.

Name: _____ Date: _____

Key Idea

A **literal equation** has more than one letter. To "solve for" one letter, get it alone on one side.

- Treat every *other* letter exactly like a number.
- Use the same moves as always: undo $+/-$ first, then undo $\times/\div$.
- **Ask yourself:** "If these letters were numbers, what would I do?" Then do it with the letters.

Common Mistake

Undo *everything* attached to your letter. From $d = rt$, dividing by r gives $t = \dfrac{d}{r}$, **not** $t = d - r$ (rt means multiply, so undo by dividing).

In a fraction answer, divide the *whole* side: from $ax = c - b$, $x = \dfrac{c-b}{a}$, **not** $c - \dfrac{b}{a}$.

Type 1 One Step: Add or Subtract

Watch Me — Worked Example

Solve $x + a = b$ for x.

Treat every letter except x like a number. If it were $x + 3 = 10$, you would subtract 3.

Step 1 Subtract a from both sides $x + a - a = b - a$

Answer $\boldsymbol{x = b - a}$

1. Solve $x + y = 10$ for x.

$x =$ _____

2. Solve $a + b = c$ for a.

$a =$ _____

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3. Solve $m - n = p$ for m.

$m =$ _____

4. Solve $y - 4 = k$ for y.

$y =$ _____

5. Solve $d + e = f$ for e.

$e =$ _____

6. Solve $t - s = r$ for t.

$t =$ _____

Type 2 One Step: Multiply or Divide

Watch Me — Worked Example

Solve $d = rt$ for t.

rt means $r \cdot t$, so t is being *multiplied* by r. Undo it by dividing.

Step 1 Divide both sides by r $\quad \dfrac{d}{r} = \dfrac{rt}{r}$

Answer $\quad \boldsymbol{t = \dfrac{d}{r}}$

7. Solve $A = lw$ for w.

$w =$ _____

8. Solve $V = Bh$ for h.

$h =$ _____

9. Solve $F = ma$ for a.

$a =$ _____

10. Solve $C = \pi d$ for d.

$d =$ _____

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11. Solve $\frac{a}{b} = c$ for a.

$a =$ _____

12. Solve $\frac{y}{4} = k$ for y.

$y =$ _____

Type 3 Two Steps

Watch Me — Worked Example

Solve $y = mx + b$ for x.

Undo in reverse order: first undo the $+b$, then undo the $\times m$.

Step 1 Subtract b from both sides $y - b = mx$

Step 2 Divide both sides by m $\frac{y - b}{m} = x$

Answer $\boldsymbol{x = \frac{y - b}{m}}$

13. Solve $ax + b = c$ for x.

$x =$ _____

14. Solve $P = 2l + 2w$ for l.

$l =$ _____

15. Solve $y = 3x - 5$ for x.

$x =$ _____

16. Solve $2x + y = 8$ for y.

$y =$ _____

17. Solve $5x - y = 10$ for y.

$y =$ _____

18. Solve $ax - c = d$ for x.

$x =$ _____

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Type 4 Parentheses

Watch Me — Worked Example

Solve $P = 2(l + w)$ **for** w**.**

The quickest way: divide by the number outside the parentheses first.

Step 1 Divide both sides by 2 $\quad \dfrac{P}{2} = l + w$

Step 2 Subtract l from both sides $\quad \dfrac{P}{2} - l = w$

Answer $\quad \boldsymbol{w = \dfrac{P}{2} - l}$

19. Solve $a(x + b) = c$ for x.

$x =$ _____

20. Solve $3(x + y) = z$ for x.

$x =$ _____

21. Solve $2(a - b) = c$ for a.

$a =$ _____

22. Solve $k(m + n) = p$ for m.

$m =$ _____

23. Solve $4(x - y) = t$ for x.

$x =$ _____

24. Solve $2(x + 5) = y$ for x.

$x =$ _____

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Type 5 Fractions and Formulas

Watch Me — Worked Example

Solve $A = \frac{1}{2}bh$ for h.

Clear the fraction first by multiplying by its denominator.

Step 1 Multiply both sides by 2 $\quad 2A = bh$

Step 2 Divide both sides by b $\quad \dfrac{2A}{b} = h$

Answer $\quad \boldsymbol{h = \dfrac{2A}{b}}$

25. Solve $A = \frac{1}{2}bh$ for b.

$b =$ _____

26. Solve $\dfrac{x+y}{2} = m$ for x.

$x =$ _____

27. Solve $C = 2\pi r$ for r.

$r =$ _____

28. Solve $I = Prt$ for P.

$P =$ _____

29. Solve $y = \dfrac{x-3}{2}$ for x.

$x =$ _____

30. Solve $V = lwh$ for h.

$h =$ _____

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